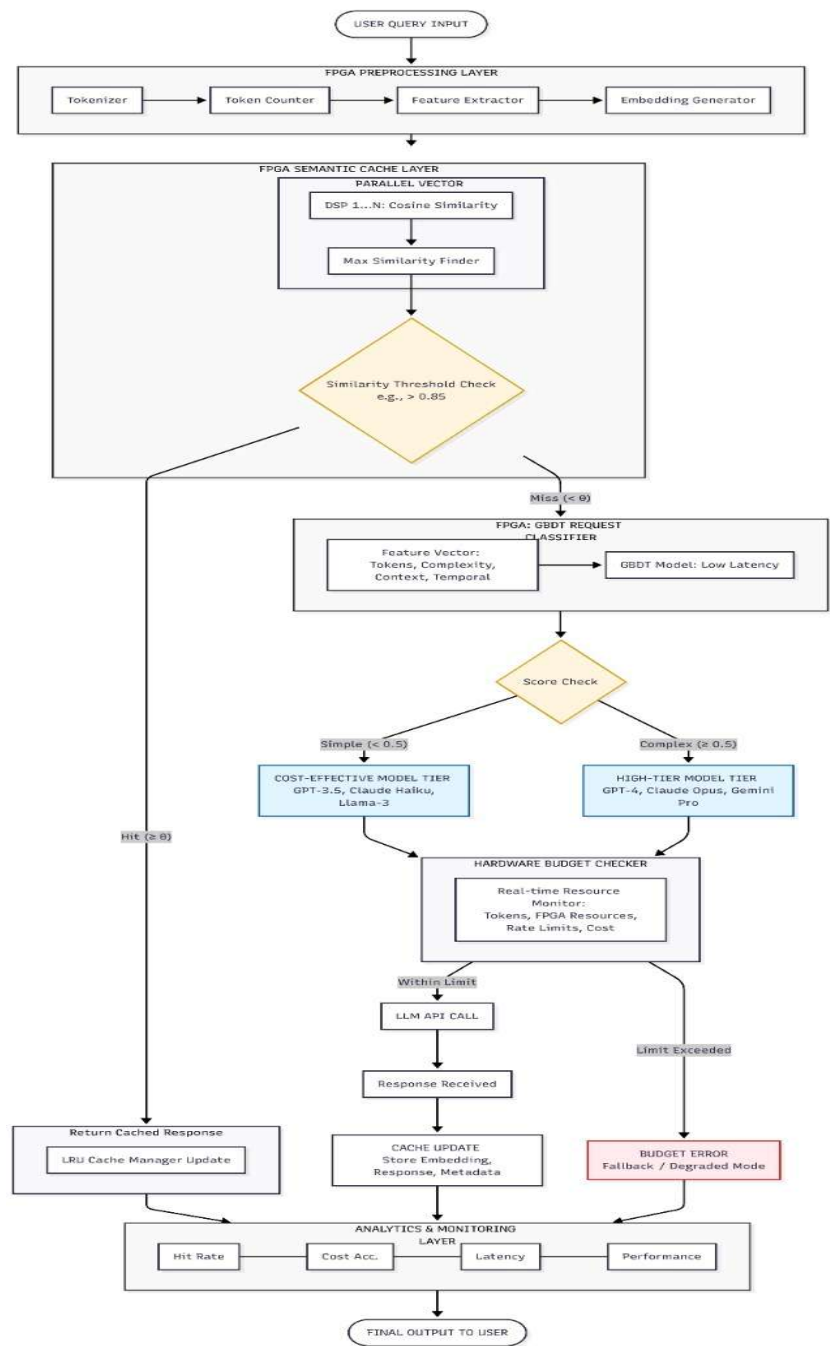


SYNOPSIS : MAJOR PROJECT 2025-26	
PROJECT NAME(Group Name)	LLM-OptimEdge (FPGA-Accelerated Semantic Caching & Intelligent Routing) (IoSec)
ELECTRONICS AND TELECOMMUNICATION ENGINEERING	
 Vivekanand Education Society's Institute of Technology	
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SEM/Year/CAY	VI-VII/TE-BE/2025-26
Problem Statement (Initial Goal)	The deployment of Large Language Models (LLMs) faces a "cost-per-token" barrier where high API expenses and computational latency hinder scalability. Conventional systems suffer from redundant API calls for similar queries and inefficient routing where expensive, high-parameter models are used for simple tasks. Existing software-based caching and routing often introduce significant latency overhead, especially when processing high-dimensional vector embeddings.
OBJECTIVE(s)	• Hardware-Accelerated Semantic Caching: Implement a vector similarity search engine on FPGA to identify and reuse previous LLM responses, avoiding redundant API costs. • Intelligent Request Routing: Develop a hardware-based classifier using Gradient Boosted Decision Trees (GBDT) to route queries to appropriate model tiers (e.g., GPT-4 vs. GPT-3.5) based on complexity. • Token & Budget Management: Design a real-time hardware token counter and budget checker to enforce hard limits on API spending. • Latency & Power Optimization: Achieve sub-millisecond similarity matching and reduce the energy-per-query compared to standard CPU-based middleware.

SDGs to be mapped	• SDG 9: Industry, Innovation, and Infrastructure (Advancing AI hardware infrastructure). • SDG 12: Responsible Consumption and Production (Reducing energy waste in data centers through hardware efficiency).
Introduction: As generative AI becomes integral to vocational and technical training, the operational cost of LLM APIs has become a critical constraint. This project proposes a hardware-software co-design approach to optimize LLM inference costs. By leveraging the parallel processing capabilities of FPGAs, we can accelerate the "Semantic Caching" process—finding similar past queries to avoid new API calls—and "Intelligent Routing"—sending simple questions to cheaper models. This system ensures high-quality AI assistance remains economically sustainable. Description: The system integrates Internet of Things (IoT) principles with FPGA-based machine learning acceleration. The architecture consists of a middleware layer where incoming user queries are first processed by the FPGA. • Semantic Cache Layer: The FPGA implements a vector similarity engine using DSP blocks to perform high-speed Cosine Similarity or Nearest Neighbor search against a local database of embeddings. • ML-Driven Router: Inspired by EDA optimization techniques, a GBDT classifier predicts the "Quality of Results" (QoR) for a query. If a query is classified as "Simple," it is routed to a low-cost model; "Complex" queries are sent to high-tier models. • Active Learning Optimization: The system employs an active learning loop to adjust similarity thresholds and cache sizes dynamically, ensuring the best trade-off between cost savings and response accuracy. This approach directly supports the "Skill India" mission by providing reliable, low-cost AI training infrastructure.	

Block Diagram of LLM-OptimEdge (FPGA-Accelerated Semantic Caching & Intelligent Routing):-



Mentor Name & Signature with date